

Question	Marking guidance	Mark	Comments
03.1	States diameter = 1.89 (mm) or radius = 0.945 (mm)	1	Convincing treatment of powers of 10 necessary for 2 nd mark
	Area = $\pi \times (0.945 \times 10^{-3})^2 = 2.81 \times 10^{-6} \text{ m}^2$ (must see at least 3 sig. fig.)	1	
03.2	$R = \frac{1.7 \times 10^{-8} \times 0.85}{2.81 \times 10^{-6}}$ $= 5.1 - 5.2 \times 10^{-3} \Omega$ 2 sig. fig.	1	1 st mark is for a correct substitution or rearrangement
		1	Correct answer gets first 2 marks
		1	3 rd mark is stand alone

Question	Marking guidance	Mark	Comments
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16	C
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Question	Marking guidance	Mark	Comments
05.1	not a straight line (through the origin) or V not proportional to I or resistance is increasing or changing or gradient not constant ✓	1	
05.2	MAX 3: electrons collide with lattice / ions / atoms ✓ idea that there is a transfer of energy between electron and lattice ion ✓ vibration of ions/atoms increases ✓ resistance / temperature increases ✓	3	Accept frequency or amplitude or just vibration
05.3	Two correct read-offs $V_{\text{lamp}} = 2.65$ to 2.75 V; $V_{\text{diode}} = 0.63$ to 0.64 V ✓ both values added ✓	2	Expect 3.28 to 3.39 correct answer (without working) gets both marks

Question	Marking guidance	Mark	Comments
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Question	Answers	Additional comments/Guidelines	Mark	ID details
01.1	The resistance between opposite faces of a unit cube (of the material) ✓	Accept resistance x <u>cross-sectional</u> area divided by length. Do not accept equation in symbols only.	1	E

Question	Answers	Additional comments/Guidelines	Mark	ID details
01.2	Use of $R = \frac{\rho l}{A}$ ✓ 0.63 (m) ✓	Substitution or rearrangement If diameter used for radius, give 1 mark max provided answer given (2.5 m) Look for 4.52×10^{-8} for area Do not accept 0.62 (m)	2	E
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	ID details
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	7.67×10^{-4} (s) OR 7.7×10^{-4} (s) ✓	Penalise 1 sf	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	6.8×10^7 (V) ✓	Penalise 1 sf	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Use of $E = Pt$ leading to 1.2(3) (s) ✓	Penalise 1 sf	1	AO2
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Marking guidance	Mark	Comments
05.1	0 or zero ✓	1	Do not credit ambiguous answers

Question	Marking guidance	Mark	Comments
05.2	Arrow drawn from any excited level ($n > 1$) down to $n = 1$ level ✓	1	Do not accept line from ionization level

Question	Marking guidance	Mark	Comments
05.3	Uses $hf = E_1 - E_2$ ✓ Multiplies by $1.6 (\times 10^{-19})$ ✓ Uses the $(3.40 - 1.51 =)$ 1.89 eV transition ✓ Correct answer for any transition shown in additional comments ✓	4	for any transition look for 1.6×10^{-19} in the equation or for 3.024×10^{-19} (J) for completely correct answers expect to see 4.56×10^{14} (Hz) but accept 7.29×10^{14} (Hz) or 6.90×10^{14} (Hz) or 6.15×10^{14} (Hz) for MP4 Only 4.56×10^{14} (Hz) answer gets 4 marks Penalise POT and rounding errors in MP4 (calculator answer is 4.56109) Allow answer with 2 sf. Condone answer of 4 or more sf.

Total		6
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Question	Marking guidance	Mark	Comments
09.1	(The potential differences are) equal ✓	1	OWTTE

Question	Marking guidance	Mark	Comments
09.2	Uses half the range: 0.01 ✓ Calculates the mean: 1.18 ✓ % uncertainty = $\frac{100 \times \text{their range}}{\text{their mean}}$ % ✓	3	Accept 1.178 for MP2 Expect to see 0.9% Condone 2 sf answer (0.85%) Allow MAX 2 if using data for x and y: Calculates mean of x and y values (539 and 457) ✓ Uses half the range (2) to determine % uncertainty of x and y And adds them (0.37 + 0.44 = 0.8(1)%) ✓ Only accept 1 or 2 sf final answer.

Question	Marking guidance	Mark	Comments
09.3	Point plotted at (1.18, 4.7) with an error bar that is 4 to 4½ small squares ✓	1	Allow +/- half a grid square for point

Question	Marking guidance	Mark	Comments
09.4	Well drawn line of best fit ✓	1	The line should pass through all the error bars with an even scatter of points either side. Candidates may have this mark even if they do not plot the final point. Expect an S intercept of 2.02 to 2.12 Do not accept multiple, thick or unruled lines.

Question	Marking guidance	Mark	Comments
09.5	Large triangle drawn on line or with values shown clearly in an acceptable way on the graph ✓ Correct data extraction from the graph ✓ Calculates a gradient to determine R that rounds to 3.7 to 3.9 Ω ✓	3	Triangle should be formed on more than half their line. 2 or 3 sf only

Total		9
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11	B	$\text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$
		ν

12	A	$\frac{V_s}{I_s}$ infinite
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13	D	Q and S.
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14	A	infinite zero
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	Use of $E = \frac{hc}{\lambda}$ ✓ Use of $P = NE$ with their value of E ✓ Correct answer (3.3×10^{16}) ✓		3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Value of R_L from graph = 30 kΩ ✓ Use of potential divider equation to give their R_L divided by 2 ✓	Allow 28 to 32 kΩ accept ratios	2	1 × AO3 1 × AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	(at greater light levels) R_L decreases ✓ (therefore must) reduce $R_{\text{variable resistor}}$ ✓ (So that) the ratio of the resistances returns to original value / so that V_I returns to its original value ✓	Allow MP3 for current discussion leading to V_I returning to original value.	3	3 × AO3
Total			8	

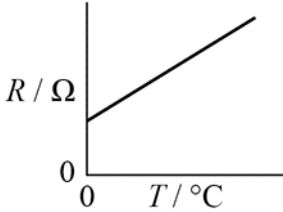
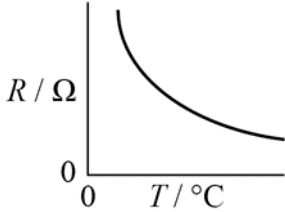
Question	Answers	Additional comments/Guidelines	Mark	AO
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16	B	increases increases	AO1
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17	D	8 <i>R</i>	AO2
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18	C	1600 W	AO2
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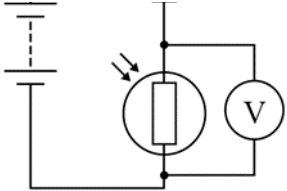
21	D	C Ω s	AO3
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22	B	 	AO1
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12	D	16R	
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13	C	2.6×10^{21}	
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20	B	23 V	
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21	A	

Question	Answers	Additional comments/Guidelines	Mark	AO
04	read-off voltage = 0.144 V (correct to half small square) ✓ Use of $R = \frac{V}{I}$ leading to $R = 180\ (\Omega)$ ✓	Do not allow 0.142 V Allow 1 mark total for full calculation done from incorrect V (outside the tolerance)	2	AO3 AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	(Use of $P = \frac{V^2}{R}$ leading to) 2900 (W) ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Power/energy/heat dissipated in nichrome/in the water must be much greater (than power/energy/heat dissipated in copper wires) ✓₁ (in series) the voltage is shared in the ratio of resistances ✓₂ Voltage must be much higher across the nichrome (than the copper) since $P = IV$ ✓₃ OR the current is the same in the copper and the nichrome ✓₂ $P = I^2R$ so resistance of nichrome must be higher ✓₃	Allow 2 marks max for: compares local mains voltage with quoted voltage across nichrome wire ✓ current must be the same in nichrome and cable ✓ If no other mark awarded, allow one mark for to avoid energy/power loss in cable or to reduce overheating in copper cable	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	Use of $R = \frac{\rho L}{A}$ OR evaluation or correct expression of A ✓ 1.3 (m) ✓ (1.29 m to 3sf)	$A = 8.0 \times 10^{-8} \text{ m}^2$ Allow POT error in MP1 Do not allow 1.30m (rounding error)	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	Power consumption decreases (for the first 5 ms) then is constant thereafter ✓ Because: R increases as T increases (for the first 5 ms) ✓ V stays the same AND statement of $P = \frac{V^2}{R}$ or equiv ✓	Full marks can be awarded for a correct discussion that takes into account voltage varying slightly due to nichrome:copper potential divider	3	AO2
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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14	C	0.78 A	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02	Max ✓✓ Uses area of graph OR states that area = Q Determines total area (in A h) Converts A h to C 7.8×10^3 (C) ✓	Total area = 2.172 A h Max 2 for counting squares method: accept 22 ± 0.5 squares of 0.1 A h. Calculator value is 7819.2 C	3	1 × AO1 2 × AO3
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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14	C	$30\ \Omega$	AO3
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12	D	60 Ω	AO3
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13	D	1.3 W	AO1

14	A	$\frac{V}{R}$	0	V	AO1
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15	D	6 Ω	AO2
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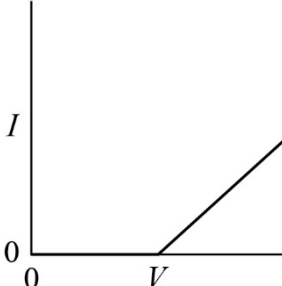
16	A	$0.87I$ $0.87P$	AO1
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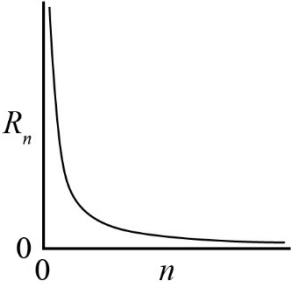
Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	C ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	1.6×10^4 ✓	Ignore unit	1	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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12	A	cool the superconducting wires.	AO1
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13	B		AO3
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14	D		AO3
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